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PROVING THE AGE OF A PALEOMAGNETIC POLE: THE CASE OF THE NTONYA RING STRUCTURE, MALAWI

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Abstract. Geological application of paleomagnetic data generally relies on equating radiometric or biostratigraphic age with that of the characteristic remanence. Yet the standard of proof of paleomagnetic age is often unsatisfactory and is extremely difficult to achieve where contact tests or conglomerate tests are unavailable. In this study of the Ntonya igneous ring complex the thermoremanent (TRM) origin of one component of natural remanent magnetization (NRM) is confirmed by paleointensity determination and its carrier is identified mineralogically, and the age of this TRM is estimated by extrapolation from $^{40}Ar/^{39}Ar$ mineral ages. The characteristic remanence in the Ntonya Ring Structure, which was determined by Briden (1968) as D = 311° I = +43°, α_{95} = 1.9°, pole 28°N 15°W, is shown to be a TRM in single-domain/pseudo-single-domain magnetite, acquired in a field of approximately half the present geomagnetic field strength. The $^{40}Ar/^{39}Ar$ mineral ages (biotites 471.5 ± 7 Ma; hornblendes 510 ± 7 Ma) and the high precision of the paleomagnetic directions combine in a model of steady cooling (~ 4.6° m.y.$^{-1}$) that leads us to the conclusion that the thermoremanence was acquired over approximately 13 m.y., 522 ± 13 m.y. ago. In passing it is shown that the isothermal (IRM) component that dominates the remanence is carried by multidomain magnetite. The study shows how remanence age may be determined with precision in the absence of geological constraints. It makes the Ntonya pole a key point in the African (and Gondwanan) apparent polar wander path, correlating with Middle Cambrian glacial deposits in the Sahara at about 80°S paleolatitude.

Introduction

The determination of a late Precambrian/early Paleozoic pole for Africa from the Ntonya Ring Structure, Malawi, [Briden, 1968] as being located close to the present-day coast of Mauritania was intriguing when it was published both because it implied a very large amount of drift of the African continent in the Phanerozoic and because of the possibility that this position of the pole might be correlated with either the late Precambrian or early Paleozoic glacial deposits in the Saharan region. Probably because of its unusually high precision (A_{95} = 1.8°) the pole soon became regarded as a key piece of paleomagnetic data for Gondwana despite the large spread of radiometric ages and the absence of any geological field tests to constrain the age of the paleomagnetic pole. We have recently undertaken extensive rock magnetic work and $^{40}Ar/^{39}Ar$ analyses, the results of which (1) demonstrate that the characteristic magnetization is thermoremanent (TRM) carried by pure single-domain magnetite and (2) provide us with a cooling model which allows us to estimate both the age of this remanence and the period and temperature range over which it was acquired. We can now understand the high paleomagnetic precision and the spread of age determinations, which were previously puzzling.

In the course of the work we obtained a spot determination of the intensity of the Middle Cambrian paleomagnetic field and also demonstrated that the large isothermal (IRM) component that dominates the natural remanent magnetization (NRM) is carried by pure multidomain magnetite.

Previous Work

The Ntonya complex is centered about 5 km south of Zomba at 35°19'E, 15°27'S. It was the subject of mapping and petrological work by Bloomfield [1965], and there is no more recent work to our knowledge. The complex is about 7 km in diameter and consists of a central mass of quartz syenite surrounded by a ring of lineated granite in the form of a cone sheet. It is intruded into charnockitic gneisses of the late Precambrian to early Paleozoic (pan-African) Mozambique Belt [Bloomfield, 1965, 1970]. According to Cahen et al. [1984] the granulite facies metamorphism in the Mozambique Belt may be as old as 845 Ma. However, in southern Malawi, later folding, migmatization, and shearing, accompanied by granite and pegmatite intrusion, were widespread at about 630 Ma. The chemistry and mineralogy of the Ntonya complex as well as the presence of xenoliths and roof pendants of the gneissic country rock indicate that the complex was derived from at least partial melting of basement similar to its present host rocks. Absence of strong fabric within the quartz syenites and circularity of the outer ring, and in particular the absence of any fabric or elongation of the complex parallel to the regional structural and metamorphic trend, NNE-SSW, point to emplacement at a late stage in the Pan-African orogeny. The absence of chilled margins or even of finer grained syenite or granite towards the margin point to intrusion into hot, even semiplastic, rock. The fact that the contact from the lineated granite into the surrounding gneisses is gradational, may indicate that vertical displacement of the intrusion relative to its host

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is small, and hence that emplacement took place at great crustal depths at temperatures close to those at which granite melts.

The paleomagnetic study by Briden [1968] involved at least two 60-cm-deep cores being taken from each of 10 sites in the anatectic quartz syenites and one in the granite. The latter did not yield interpretable results, perhaps due to its strong mineral fabric. In the syenites, total NRMs were very strong (0.5-13 A m^{-1}) and randomly directed. Alternating field (a.f.) demagnetization (Figure 1) shows the NRM to be dominated by a low-coercivity component, randomly directed between cores, and of diminishing intensity with depth below outcrop. It is deduced to be IRM caused by lightning.

After a.f. cleaning, remanence directions at seven sites are exceptionally tightly grouped: the statistics on the site mean directions are D = 311.3°, I = +42.8°, k = 1054, α_{95} = 1.9°, and corresponding pole is 27.7°N, 15.1°W, A_{95} = 1.8°. Of the remaining sites, directions at two move towards the quoted mean direction during both a.f. and thermal demagnetization but do not reach it. At the other site, directions seem to move towards the reverse direction but do not reach it. Hence these three sites do not figure in the quoted result.

Rock Magnetic Studies

E. McClelland and J. C. Briden (manuscript in preparation, 1992) have shown that when the low-coercivity remanence carrier in these rocks is given a partial TRM (PTRM), even at temperatures hundreds of degrees below the Curie temperature, the PTRM does not fully unblock until heated close to the Curie temperature. Bol'shakov and Scherbakova [1979] have suggested that this behavior is diagnostic of multidomain grains: large grains, up to 500 μm diameter, with compositions close to pure magnetite (Ti:Fe < 0.01) are common in these rocks. Some of these appear homogeneous when examined with the electron probe microscope, often occurring adjacent to homogenous ilmenites (Figure 2a); others contain ilmenite exsolution lamellae (Figure 2b).

E. McClelland and J. C. Briden [1991] have carried out detailed thermal demagnetization/remagnetization studies [Thellier and Thellier, 1959] on the high-coercivity component in several samples from Briden's [1968] site A, i.e., on samples in which the remanence in low-coercivity grains has been removed by repeated, partial alternating field demagnetization at each step of the experiment. They have shown that PTRM imposed on these samples demagnetizes within a few degrees of the maximum temperature at which the PTRM had been imposed, i.e., blocking and unblocking are equivalent. Hence the high-coercivity remanence carrier, which carries the characteristic remanence (Figure 1), is single domain/pseudo single domain (PSD). The Curie temperature is ~580°C, indicative of almost pure magnetite. Fine grains (less than 1-15 μm) of subhedral, roughly equidimensional pure magnetite (i.e., with no detectable Ti under the electron microscope) occur in scattered patches within perthitic feldspar grains and could be the carriers of this remanence (Figure 2c), although whether the maximum size of PSD magnetite is as large as ~10 μm as proposed by Halgedahl and Fuller [1983] remains to be confirmed.

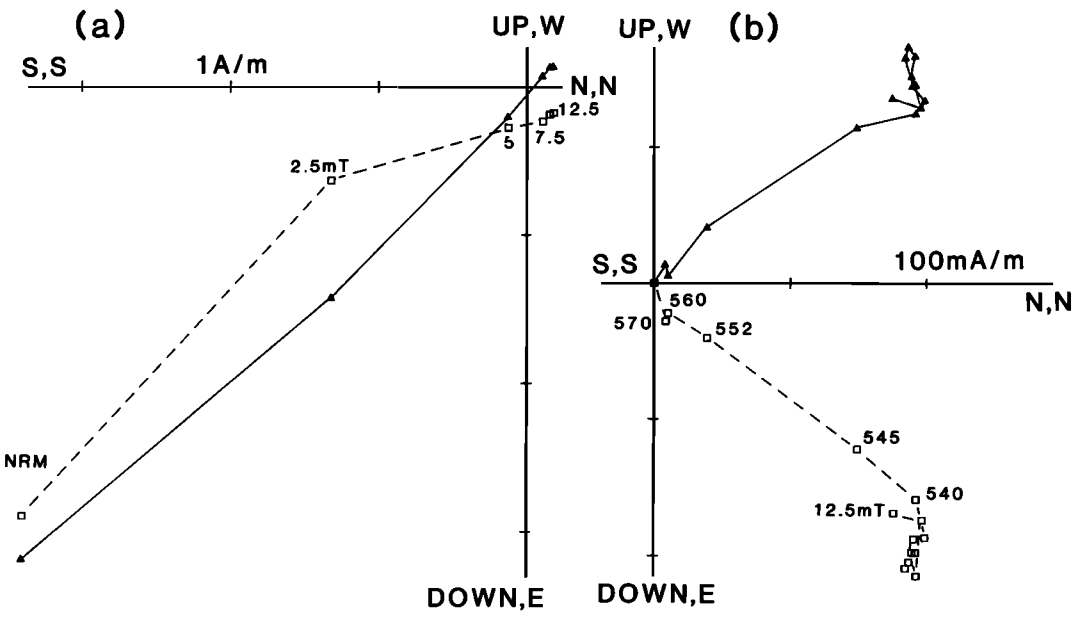


Fig. 1. Orthogonal projections of (a) partial alternating field demagnetization, followed by (b) thermal demagnetization of sample MRA 2.3 from Ntonya Ring Complex, Malawi. Solid triangles represent horizontal projection; open squares represent vertical N-S projection. The characteristic component is directed downwards to the NW and has laboratory critical blocking temperatures concentrated between 530°, and 560°C, extending to 580°C.

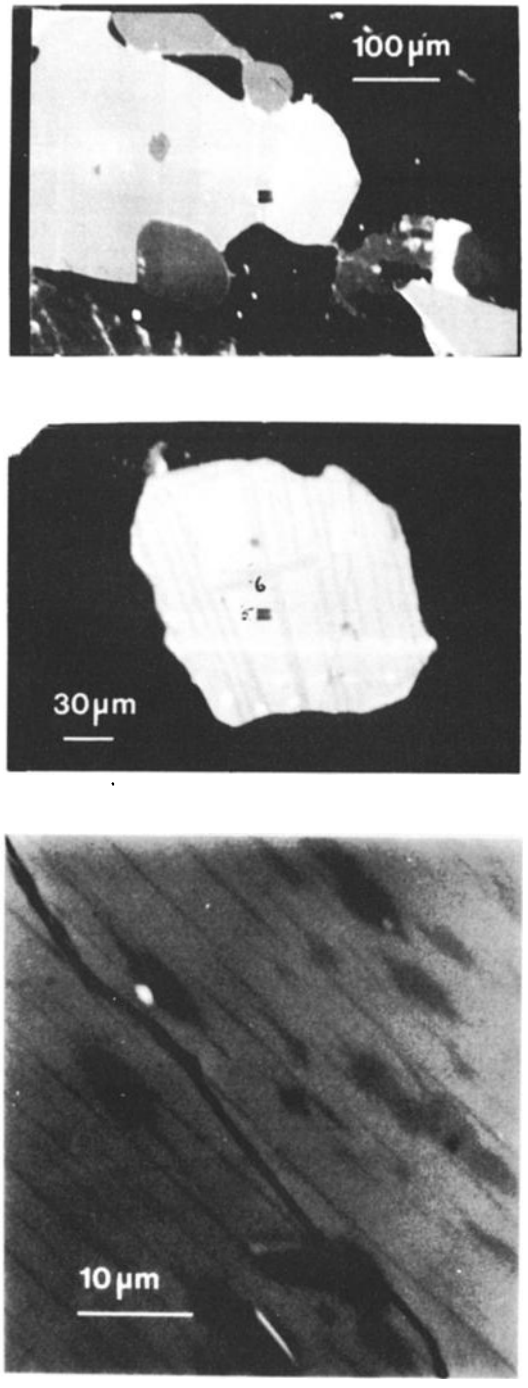


Fig. 2. Iron-titanium oxides in paleomagnetic samples: (a) composite pair of ilmenite and magnetite grains (about 400 µm) consisting of homogeneous subgrains with compositions within 1% of pure end-members (back scattered electron image); (b) subhedral magnetite (about 200 µm) with no more than 1.3% Ti, containing fine ilmenite lamellae (back scattered electron image; analyzed points are marked); (c) fine magnetite (~1.5 µm) containing no detectable Ti, in perthitic feldspar grain (scanning electron micrograph); we have analyzed much smaller similar grains but have not obtained clear enough photographs for reproduction.

The characteristic remanence has blocking temperatures concentrated in the range 530°-560°C (Figure 3a). Normalized NRM/TRM demagnetization curves are virtually identical over this temperature range (Figure 3b) strongly suggesting that this characteristic remanence is a TRM [Thellier, 1937]. NRM/TRM plots (e.g. Figure 3c) indicate acquisition in about 20 µT, i.e., approximately half the present Earth's field. This would be an underestimate if the characteristic remanence has any reversed polarity within it, but there is no evidence for this in our experimental data.

Geochronology

The K-Ar mineral ages quoted by Briden [1968], recalculated using modern decay constants [Harland et al., 1990] ranged from 482 ± 16 to 546 ± 16 Ma (biotite) and 557 ± 85 Ma (hornblende with minor pyroxene). Briden used regional geochronological arguments for slow cooling in the Mozambique orogenic belt by Cahen and Snelling [1966] to conclude that the age of stable NRM was likely to be around 600 Ma and this seemed to be reinforced by an unpublished Rb/Sr isochron of 630 ± 24 Ma by one of us (D.C.R) which was subsequently quoted by McElhinny and Briden [1971] and Piper [1972] and became accepted by most commentators but not all [e.g., McWilliams, 1981].

The Rb/Sr age estimate was derived from a regional isochron including not only two samples from the Ntonya complex but also a variety of metamorphic rocks from the Mozambique belt in the area. The drawing of such an isochron without adequate control on $^{87}Sr/^{86}Sr$ initial ratios for each contributing rock unit is no longer regarded as acceptable procedure, and this estimate should not be used in future.

The new $^{40}Ar/^{39}Ar$ analyses (Figure 4) were made on biotites and hornblendes from the paleomagnetic core samples from four of the seven localities that contributed to the paleomagnetic pole [Briden, 1968, Figure 1]. Three biotites average 471.5 ± 7 Ma, and four hornblendes average 510.0 ± 7 Ma. Closure temperatures for these minerals in slowly cooling terrains (5°C/m.y.) have been estimated as 300°C for biotite [Harrison et al., 1985] and 470°C for hornblende [Harrison, 1981]. Errors in these estimates are difficult to quantify; we have taken ±40°C error for biotite, following Harrison and McDougall [1980], and ±10°C for hornblende, since this is the range ascribable to a factor 2 variation in effective diffusion radius [Harrison, 1981].

Interpretation

If we assume a uniform rate of cooling between the times that the hornblende and biotite systems closed, then the cooling rate was about 4.6°C/m.y. All our samples from the Ntonya complex carry a single polarity stable remanence that has a narrow blocking temperature range, typically 560°-530°C. This is equivalent to natural blocking at 555°C-495°C at that cooling rate [Dodson and McClelland Brown, 1980, 1985]. This model leads us to estimate that NRM acquisition was spread over about 13 m.y. at an age of 522 ± 13 Ma (Figure 5). If cooling rate progressively decreased (as by the curved line in Figure 5), the age of acquisition

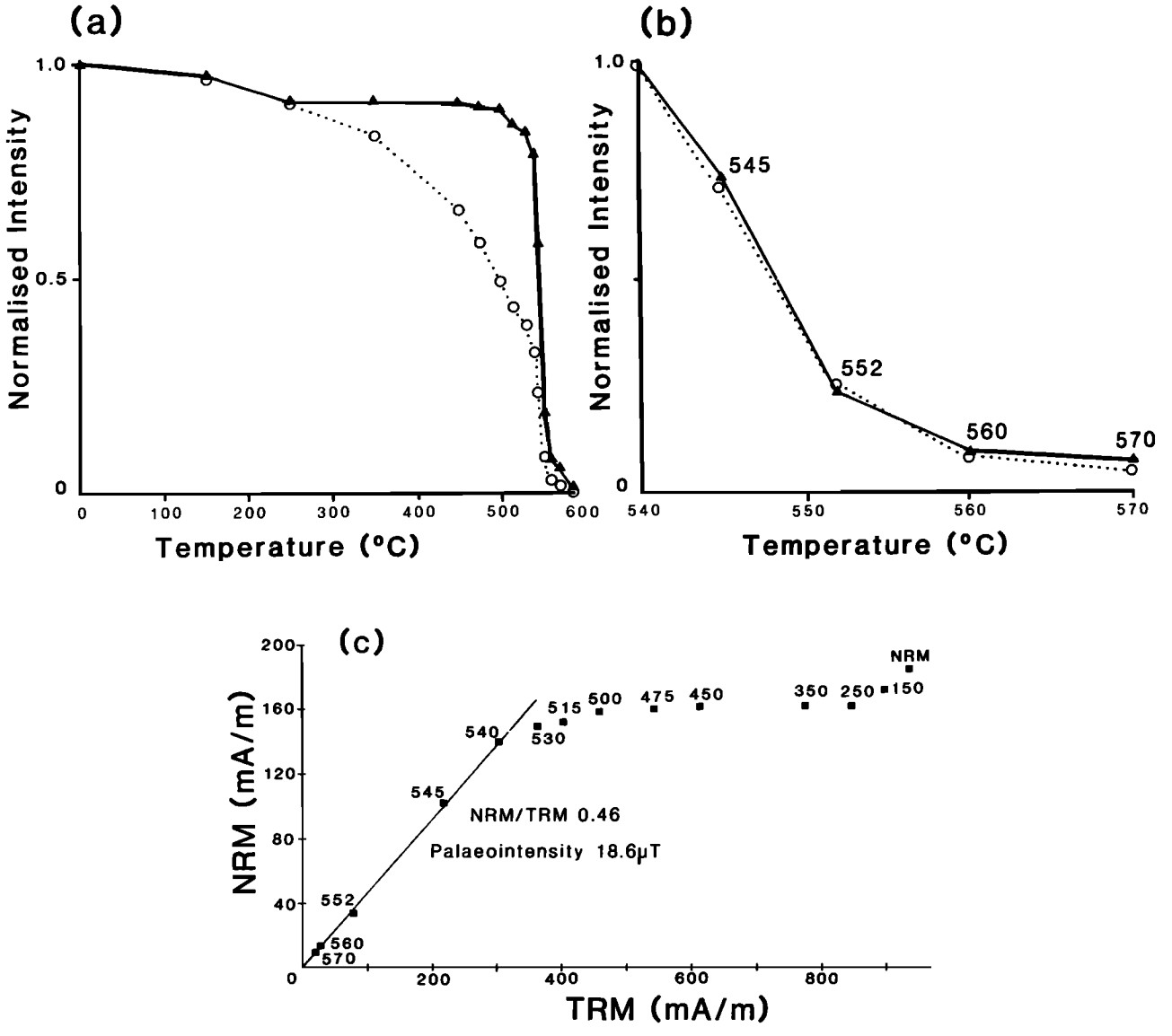


Fig 3. Normalized thermal demagnetization curves of natural remanent magnetization (NRM, solid line and triangles) and thermoremanent magnetization (TRM, dotted line and open circles) acquired by the higher coercivity grains (H_{cr} > 12.5 mT) in sample MRA 2.3 (same sample as illustrated in Figure 1). (a) Whole temperature range and (b) above 540°C. (c) Ratio NRM/TRM remaining after demagnetization of each to the indicated temperature. Points lying on the straight line are used to calculate the paleofield intensity. The effect of low-coercivity grains on all these measurements was eliminated by alternating field demagnetization at 12.5 mT; the acquisition field of the TRM was 40 μT.

would have been in the younger part of the quoted range, and the duration of acquisition would have been shorter. The present state of knowledge of the geomagnetic polarity time scale for the Cambrian is insufficient to allow useful comment on the evidence from this study, that there may have been a normal polarity chron lasting more than 10 m.y.

In the original paleomagnetic paper, Briden [1968] interpreted the high precision of the paleomagnetic directions to mean that the stable NRM had been acquired in a period of time within which secular variation had been averaged out but polar wandering was small, i.e. a few million years. The new $^{40}Ar/^{39}Ar$ data, coupled with the demonstration of TRM, reinforce this interpretation.

The cooling rates indicated by these isotopic data are considerably slower than would be expected for magmatic cooling of an intrusion of the size of the Ntonya complex, emplaced in the upper crust. The petrogenetic and structural evidence summarized above [Bloomfield, 1965] points to the conclusion that these rocks were emplaced and cooled in the lower crust. Thus the cooling that is recorded in the paleomagnetic and radiometric data is likely to be due to postorogenic exhumation.

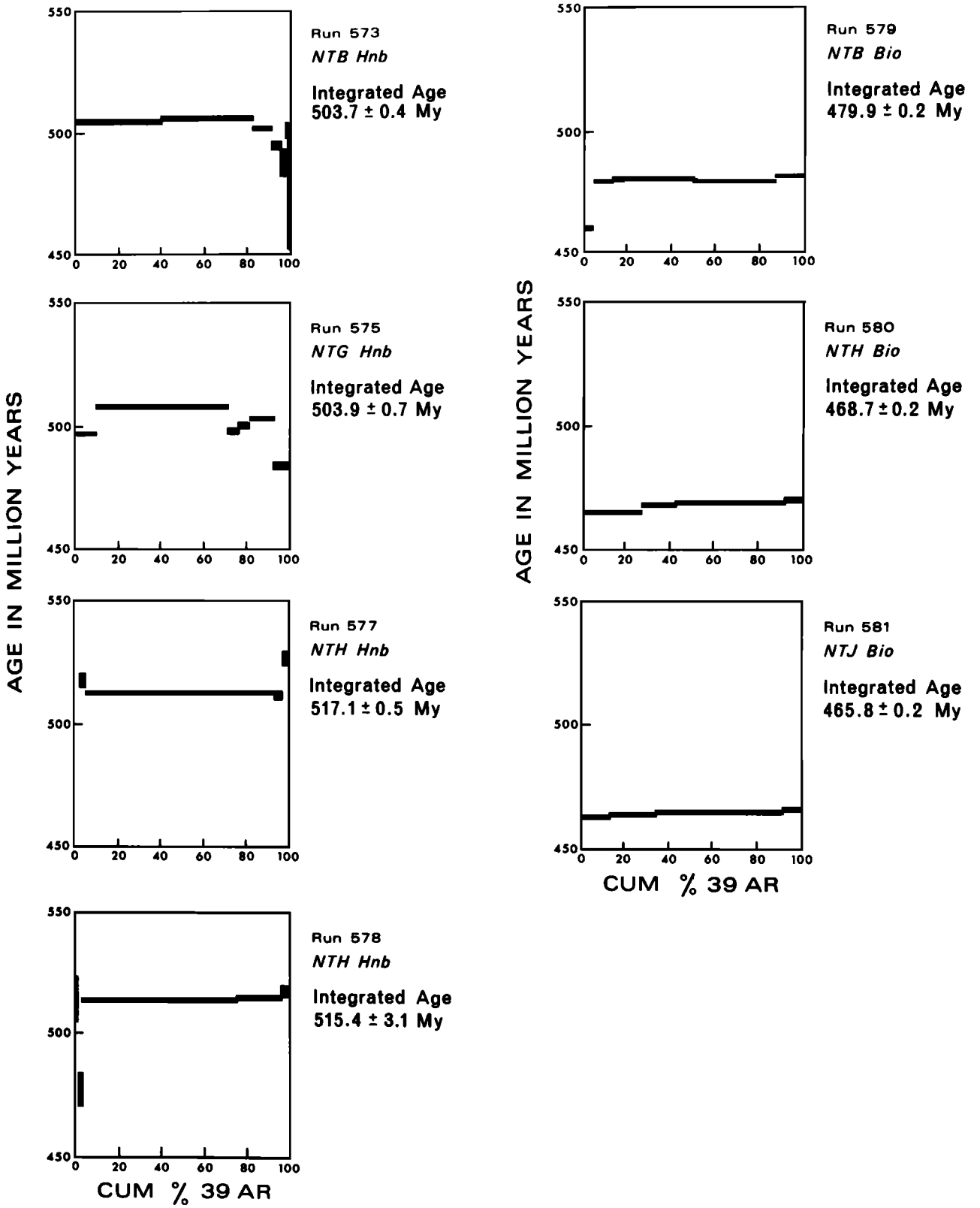


Fig. 4. The $^{40}Ar/^{39}Ar$ age determinations by step heating of irradiated samples. Calculated age at each step is plotted against cumulative percent release of ^{39}Ar. Analytical errors (2σ) are depicted but in most cases are within the thickness of the drawn lines. The measurement of the last 0.5-0.2% of Ar to be released in three of the hornblende analyses was subject to large 2σ errors and has been omitted from the age calculation. Full analytical details are given by Parsons et al. [1988]. The neutron flux (J) monitors used were the amphibole standards FY12a Hnb, HB - 3Gr Hnb, concentration as quoted by Roddick [1983].

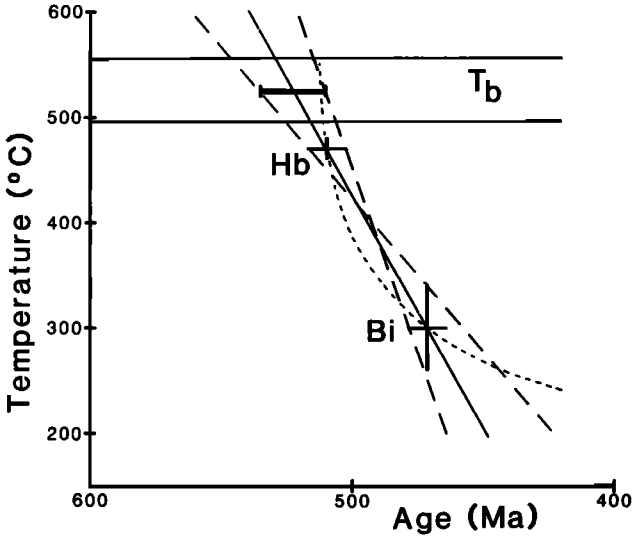


Fig. 5. Cooling profile for Ntonya Ring Structure controlled by $^{40}Ar/^{39}Ar$ closure age and temperature in hornblendes (Hb) and biotites (Bi) (see Figure 4 and text). The blocking temperature range of the characteristic remanence, recalculated for a cooling rate of 4.6°C/m.y, is indicated by the two horizontal lines marked T_B. The age of thermoremanence is established by extrapolating extreme cooling curves to intersect the mean critical blocking isotherm of the remanence. The age determinations are quoted with ±2σ errors, so our estimates of error on the remanence age are likely to be greater than 2σ. The dotted line shows an alternative cooling model (see text).

Implications

Tectonic disturbance since acquisition of TRM magnetization is likely to have been small because the granite cone sheet at its contacts with both the central syenite and the host gneiss dips consistently at ~35° towards the center of the ring. Moreover, the outcrop is approximately circular, two closely related complexes to the west are of near perfect ring form, and Karroo (about 200 Ma) and Chilwa (about 100 Ma) dykes cutting the basement are vertical [Bloomfield, 1965]. Hence the very high precision with which the magnetization direction is determined may be a reasonable indication of the accuracy of the paleomagnetic pole in spite of the many kilometers of uplift and exhumation that the presently exposed rocks have undergone since they acquired their TRM.

This study shows that the pole relates to Middle Cambrian time [Harland et al., 1990] and correlates more closely with those Saharan glacials that are now regarded as of that age, e.g., Série Pourprée in the Western Hoggar [Caby and Fabre, 1981], which are now inferred to have formed at ~80°S, rather than with the late Precambrian glacial deposits in the region. It is a key point on the early Paleozoic apparent polar wander (APW) track for Africa. It does not lie on the supposed late Proterozoic APW track [e.g., McElhinny and Embleton, 1976], and the evidence for that track is thus severely weakened.

The broader importance of this study lies in its demonstration of mode of origin, and hence age, of remanence in the absence of geological controls such as contact or fold tests.

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